
AUTONOMOUS
SYSTEM ENGINEER

PORTFOLIO

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EDUCATIONAL JOURNEY



FEB/2025

DANMARKS TEKNISKE UNIVERSITET
MASTER OF SCIENCE IN AUTONOMOUS SYSTEM



OCT/2017 - AUG/2021

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
BACHELOR OF ELECTRICAL ENGINEERING (ELECTRONICS)



FINAL YEAR PROJECT SMART ELECTRIC VEHICLE

- *Electrified internal combustion engine vehicle into an electric vehicle by integrating LiFePo4 batteries, electric motor and differential to the rear axle of the vehicle.*
- *BMS was integrated to monitor SOC and real time battery status .*
- *Smart parking assistance feature was implemented using LIDAR initially which assist in reverse parking.*





EMPLOYMENT JOURNEY

APRIL/2025 - ONWARDS

TRACE

EMBEDDED SYSTEMS

- Developed proof of concept (PoC) BLE firmware for Smart Ring in C which was deployed on nRF development boards in initial phase.
- Designed 4-layer PCBs nRF based SOC for patch prototyping .
- Integrated mobile application with nRF5 push notifications for smart ring and patches to alert user.
- Collaborated with product designer for designing compact product.



APRIL/2024 - JAN/2025

ENERGY N AUTOMATION PAKISTAN (PVT.) LTD

EMBEDDED SYSTEMS

- Developed IOT Gateway device HW/SW for Smart Grid and Data Center monitoring.
- Designed Smart Solar inverter analyzer for acquiring real time data.
- Optimized firmware of smart grid monitoring device which enhance the overall device performance.
- Led the team of technicians for quality checks of product which reduces the error and increase in product reliability.



NOV/2022 - MAR/2024

MONIT

ASSISTANT RESEARCH AND DEVELOPMENT ENGINEER

- Designed Smart Safety Helmet BLE - device HW/SW for bike riders which provides alert and real-time data, enhanced safety precautions through accident detection, and emergency notifications.
- Implemented Facial Recognition device for employee attendance using AI based micro-controller.
- Designed HW/SW of CAN data acquisition device for heavy duty vehicles for the detection of harsh braking through out the total trip of the vehicle.



DEC/2021 - NOV/2022

PAKISTAN AIR-FORCE 118SED UNIT

HARDWARE DESIGN ENGINEER

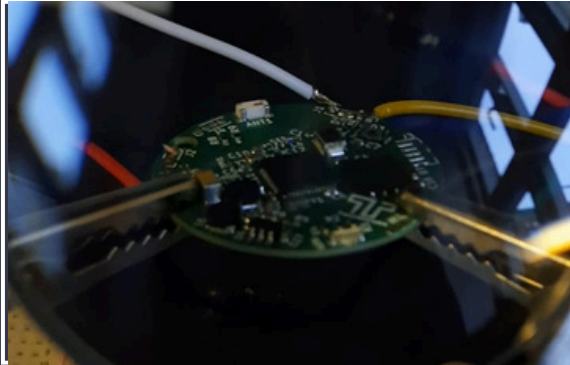
- Designed and supervised the aircraft simulator of wide body aircraft which reduces the pilot training cost.
- Integrated National Instruments Data Acquisition (NI-DAQ) hardware with a simulator, interfacing with avionics and flight dynamics modules team.
- Troubleshoot HW/SW issues in the flight training device.





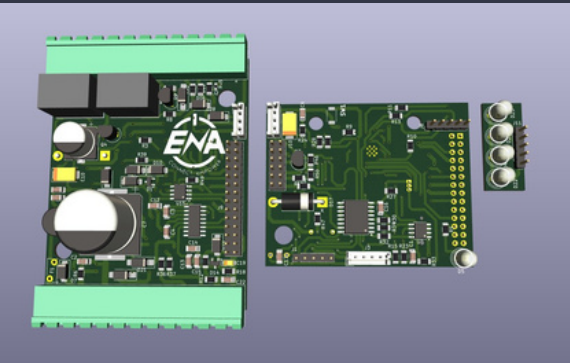
PROJECTS

PATCHES



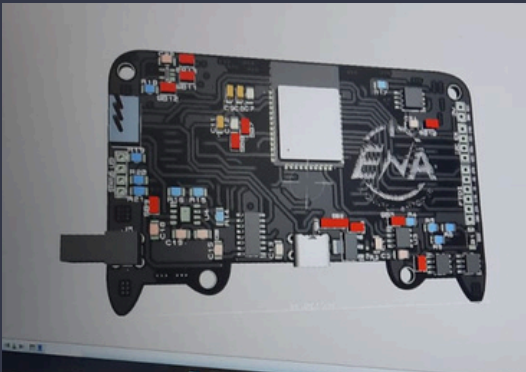
This is Patch which consist of 4-layers. It is powered by a coin cell battery and it transmits its data via BLE to Smart Ring. The purpose of patch is that it is attached to any personal belonging in-order to alert user before its left out or forgotten

IOT GATEWAY



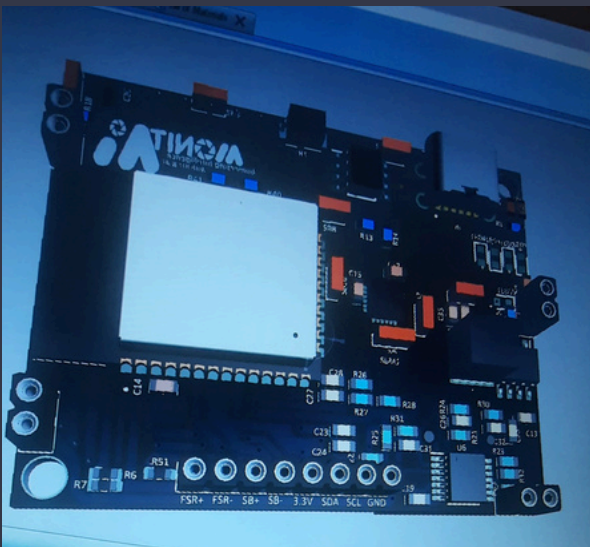
IOT-GATEWAY 2-layer PCB using KiCad software, interfacing it with an ESP32 micro controller. The PCB integrated a Real-Time Clock (RTC), RS-485, RS-232 communication along with I2C LCD display.

SMART SOLAR INVERTER ANALYZER



SSIA 2-layer PCB HW/SW. This device consist of ESP32 micro-controller interface with Real-Time Clock (RTC), LORA WAN (RF Chip) and RS232 communication IC for interfacing with inverter RS232 and gather real-time data from its port and post it to the cloud server.

SMART HELMET



Smart Helmet device 2-layer PCB HW/SW. This device consist of ESP32 micro-controller interface with MPU-6050, BLE SOC, Real-Time Clock (RTC), FSR and LIDAR sensor. This device also detect whether the helmet is put on by the rider or faking it by putting piece of cloth in the helmet. Its main purpose is to detect collision or any incident and alert it to rider company also generate SOS to near by emergency units.

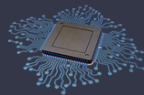


SKILLS

TECHNICAL SKILLS



ESP32/STM32
HW/SW



PCB Designing
KICAD



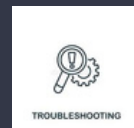
C/C++
Stm32 HAL



Data Acquisition



Micro-Python
For AI boards



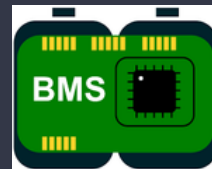
Troubleshooting
PCBs



Communication
Protocols

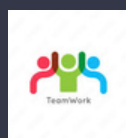


IOT & Web
Integration

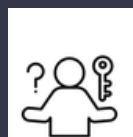


Battery Management
System and Testing

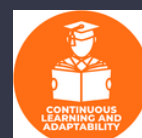
INTERPERSONAL SKILLS



Collaborative
Leadership



Analytical Problem-
Solving



Agile
Learner

TAK FOR JERES OPMÆRKSOMHED!

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